

SOLAR: A Stepwise Sleep Prediction Framework via Long-Term Anchoring and Refinement

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Abstract— Existing lifelog-based sleep prediction methods use multimodal smartphone and wearable data for daily sleep prediction but mainly rely on daily observations or short-term statistics, limiting their ability to capture long-term states. Furthermore, they often encode long-term states and day-specific variations in a single representation, making it difficult to distinguish their respective effects on sleep outcomes. To address these limitations, we propose *SOLAR*, a stepwise anchor-and-refine framework. We first construct long-term personalized features from accumulated behavioral history to capture each individual’s long-term states. The framework then generates target-specific anchors and refines them using day-specific sensor states, variations from personal long-term states, and temporal-neighbor history. Experimental results show that *SOLAR* outperforms other baseline methods.

I. INTRODUCTION

Sleep is a fundamental factor in physical and mental health, as well as daily functioning [1]. Recent advances in wearable devices enable continuous collection of sleep patterns and related physiological and behavioral data in real-world settings. At the same time, sleep patterns and their relationships with physiological and behavioral factors vary considerably across individuals, emphasizing the importance of personalized modeling. Together, these factors have driven growing research interest in predicting sleep-related targets, such as perceived sleep quality and objective sleep metrics, using multimodal personal lifelog data.

However, existing lifelog-based sleep prediction studies have two key limitations. **First**, most studies primarily rely on short-term statistics, and thus often fail to capture individuals’ long-term behavioral and physiological states. As shown in Figure 1, the 28-day rolling mean of heart rate shows higher absolute correlation coefficients than the 3-day rolling mean across multiple sleep prediction targets. This suggests that long-term heart rate patterns have a greater influence on sleep prediction than short-term patterns. **Second**, existing lifelog-based approaches that rely on long-term states still have limitations in effectively capturing the characteristics of sleep outcomes because they do not separately model an individual’s long-term state and day-specific variations [2].

To address these limitations, we propose *SOLAR* (A Stepwise Sleep Prediction Framework via Long-Term Anchoring and Refinement), a framework that explicitly separates long-term states from day-specific variation. We construct long-term features from behavioral history and individual patterns,

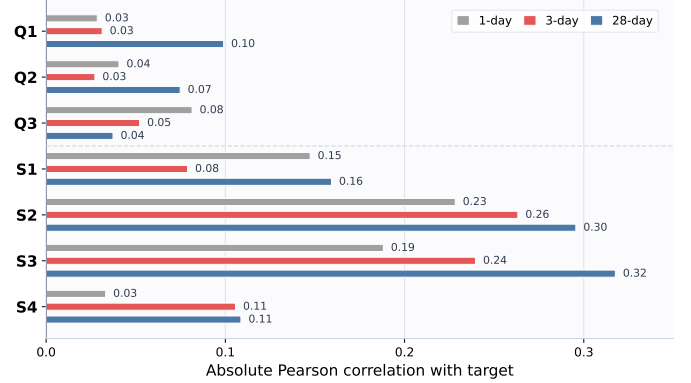


Fig. 1. Absolute correlation patterns between target variables and heart rate features computed using 1-day, 3-day, and 28-day rolling means, where each rolling mean is calculated from heart rate values observed during the corresponding preceding time window.

which *SOLAR* uses to generate a target-specific anchor. The anchor is then refined using day-specific sensor states and their variations. This design addresses these limitations by comprehensively capturing an individual’s long-term state and explicitly modeling day-specific deviations from it.

To validate these features and framework, we conduct experiments on lifelog-based sleep prediction. Experimental results show that our method achieves better performance than conventional machine learning models. Ablation studies further show that incorporating long-term features improves performance and that the complete stepwise framework contributes to better sleep prediction. The contributions of this work are summarized as follows:

- We construct long-term personalized features that capture individual behavioral states, temporal trends, and day-specific variations from typical patterns.
- We propose *SOLAR*, a stepwise sleep prediction framework that first anchors predictions on long-term features from each participant’s history and then refines them using the day-specific sensor state and its variations.
- We show in experiments and ablation studies that combining long-term and day-specific variation features improves sleep prediction over baselines across targets.

II. RELATED WORK

Prior studies in lifelog-based sleep prediction have utilized multimodal data from smartphones and wearables to infer sleep metrics based on short-term behavioral contexts [3].

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However, relying primarily on short-term features limits the ability to capture long-term states and day-specific deviations.

Furthermore, although existing sleep prediction methods consider long-term behavioral states, they often combine long-term states and day-specific variations into a single representation and learn them jointly [4]. This joint modeling makes it difficult to distinguish long-term states from day-specific variations and to capture their distinct effects on sleep outcomes. Therefore, effective sleep prediction requires separately modeling long-term states and day-specific variations before integrating them for the final prediction.

III. METHOD

We present *SOLAR*, our long-term anchor-and-refine framework for lifelog-based sleep prediction. As shown in Figure 2, *SOLAR* anchors target-wise predictions using each participant’s long-term state, refines them with personalized day-specific sensor states and variations, and stabilizes them through history-aware refinement using temporal-neighbor records from the same participant.

A. Temporal and Personalized Feature Construction

Sleep-related sensor values are difficult to interpret without considering their temporal context, because their meanings depend on both recent changes and long-term patterns in each participant’s behavioral and physiological history. Therefore, multimodal lifelog data are transformed into a temporal and personalized representation that captures short-term dynamics, long-term states, and the day-specific states.

Specifically, features computed from 1–7 days of history are defined as short-term features to capture recent lifestyle changes. Features based on 14–28 days of history or historical statistics such as cumulative mean, standard deviation, and same-weekday average are defined as long-term features to represent each participant’s long-term lifestyle state. We further construct short-long contrast features that capture how much the observations deviate from the long-term state, as well as personalized features that include subject-specific history, states, deviations, and missingness history. The day-specific state, recent temporal dynamics, and long-term state form the basis for personalized anchor prediction, while day-specific variations are incorporated during residual refinement.

B. Step 1. Personalized Temporal Anchor Prediction

Sleep outcomes are not determined solely by the day-specific sensor state, but are also influenced by temporal context such as each participant’s recurring sleep tendency, lifestyle pattern, and weekday/weekend effects. Therefore, directly predicting the target from day-specific features alone may fail to capture the participant’s long-term states.

To address this, Step 1 first constructs a personalized temporal anchor that estimates a target-wise baseline probability from each participant’s long-term state. Specifically, multiple prediction sources are generated to reflect individual history, calendar patterns, subject-level temporal priors, and long-term personalized sensor information. Since useful sources can

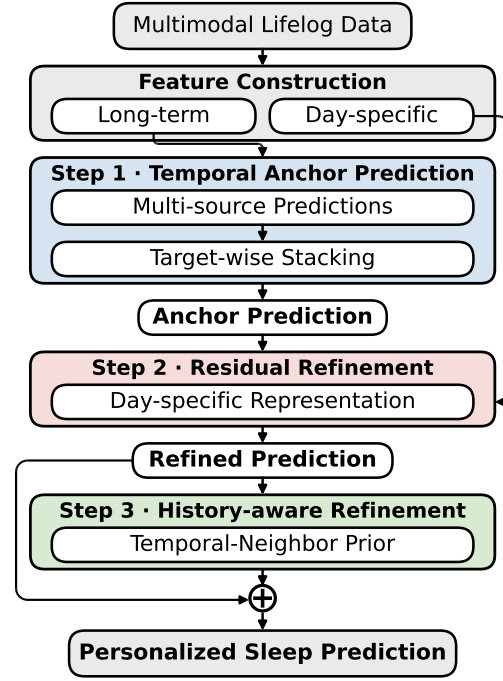


Fig. 2. *SOLAR* Overview.

differ across targets, the source predictions are selected and combined in a target-wise manner. The out-of-fold predictions are integrated through a stacking function to produce the anchor prediction for each target.

For participant i , prediction date t , and target k , the Step 1 anchor prediction is defined as:

$$\hat{p}_{i,t,k}^{(1)} = F_k \left(s_{i,t,k}^{(1)}, s_{i,t,k}^{(2)}, \dots, s_{i,t,k}^{(M_k)} \right), \quad (1)$$

where $s_{i,t,k}^{(m)}$ denotes the prediction probability produced by the m -th prediction source for target k , and M_k is the number of selected sources for target k . $F_k(\cdot)$ denotes the target-specific stacking function, and $\hat{p}_{i,t,k}^{(1)}$ is the personalized temporal anchor prediction generated in Step 1.

C. Step 2. Day-Specific Personalized Residual Refinement

Step 2 refines the personalized anchor prediction using prediction-day information. Although Step 1 estimates a stable baseline probability from long-term history and recurring temporal patterns, it may not fully capture day-specific behavioral, physiological, and environmental changes. Therefore, Step 2 uses a personalized day-specific representation to estimate the residual variation that remains unexplained by the Step 1.

Unlike a direct prediction model, this stage does not learn the target label from scratch. Instead, it learns a residual correction on top of the Step 1 anchor, using features that describe the day-specific sensor state and its deviation from the individual long-term state. In this way, Step 2 preserves the long-term state estimated in Step 1 while incorporating transient changes observed on the prediction day.

Conceptually, for participant i , prediction date t , and target k , the Step 2 prediction is defined as:

$$\hat{p}_{i,t,k}^{(2)} = \text{clip} \left(\hat{p}_{i,t,k}^{(1)} + \eta_k \hat{r}_{i,t,k}, \epsilon, 1 - \epsilon \right), \quad (2)$$

where $\hat{p}_{i,t,k}^{(1)}$ denotes the personalized anchor prediction from Step 1, and $\hat{r}_{i,t,k}$ is the residual correction derived from the personalized day-specific representation. The day-specific coefficient η_k controls the correction strength, while ϵ clips the probability away from 0 and 1. The resulting $\hat{p}_{i,t,k}^{(2)}$ is the day-specific-refined prediction.

D. Step 3. Target-specific History-aware Refinement

Although Steps 1 and 2 model each participant’s long-term state and day-specific variations, the refined predictions may still require target-specific adjustment. This is because different sleep-related targets can have different temporal dependencies and calibration behaviors. In particular, some targets are more strongly related to the recent and long-term history of the same participant, while others may not benefit from additional temporal correction.

To address this, Step 3 applies target-specific history-aware refinement based on validation performance. For targets where temporal continuity is useful, we construct a temporal-neighbor prior from the same participant’s history by assigning larger weights to observations closer to the prediction date while considering both short-term and long-term history. The prior is then used to refine the Step 2 prediction. If such refinement does not improve validation performance, the Step 2 prediction is preserved without additional correction.

Through this conservative target-wise refinement, the final prediction preserves the personalized anchor and day-specific variation modeled in the previous stages, while incorporating subject-level temporal continuity only when it is beneficial.

IV. EXPERIMENTS

A. Experimental Setup

Dataset and task. We use the public CH-2025 benchmark dataset [5], which contains about 450 daily records from ten subjects, each represented by a 2,160-dimensional multimodal feature vector from wearable devices and smartphones. The task is formulated as multi-target classification for seven sleep-related targets: Q1, Q2, Q3, S1, S2, S3, and S4. Q1–Q3 correspond to subjective sleep-related states, whereas S1–S4 represent objective sleep indicators.

Metric and baselines. Since the official test labels are not publicly available, we select the models using K-fold cross-validation and out-of-fold (OOF) predictions. We measure performance using Average Log-Loss across all seven targets and evaluate the final model on the held-out public test split through the online submission system. Lower Log-Loss indicates better calibrated probabilistic prediction. We compare our method with CatBoost [6], LightGBM [7], XGBoost [8], and SCH_CSM [9], a weighted CatBoost–LightGBM ensemble. All methods use the same preprocessing pipeline, split strategy, target-wise training scheme, and metric.

TABLE I
PERFORMANCE COMPARISON WITH TABULAR LEARNING BASELINES.

Model	Test ($\downarrow$)
CatBoost	0.6053
LightGBM	0.6316
XGBoost	0.6985
SCH_CSM (CatBoost:LightGBM = 7:3)	0.6017
SOLAR	0.5560

TABLE II
ABLATION STUDIES OF LONG-TERM FEATURES AND PIPELINE STAGES.

Configuration	Validation(OOF) ($\downarrow$)	Test ($\downarrow$)
(a) Stepwise model ablation		
SOLAR (w/o step 2 + 3)	0.5815	0.5642
SOLAR (w/o step 3)	0.5807	0.5626
SOLAR	0.5760	0.5560
(b) Long-term feature ablation		
SOLAR (w/o long-term)	0.5818	0.5739
SOLAR	0.5760	0.5560

B. Results and Analysis

We conduct experiments to answer four research questions: **RQ1** whether the proposed method improves sleep prediction over strong tabular baselines, **RQ2** whether each pipeline stage contributes complementary gains, **RQ3** whether long-term personalized features improve prediction beyond short-term observations, and **RQ4** how long-term and short-term features contribute across sleep-related targets.

Answer to RQ1. Table I compares the proposed method with tabular baselines. *SOLAR* achieves the best Log-Loss of 0.5560, outperforming CatBoost, LightGBM, XGBoost, and SCH_CSM, which obtain 0.6053, 0.6316, 0.6985, and 0.6017, respectively. This result demonstrates that our step-wise anchor-and-refine approach more effectively leverages long-term information, improving prediction performance.

Answer to RQ2. Table II, (a) reports the stepwise ablation. Removing both Step 2 and Step 3 results in 0.5642, and removing only Step 3 obtains 0.5626. Finally, the full *SOLAR* model achieves the best score of 0.5560. Experimental results show that each step of *SOLAR* contributes to performance improvements, demonstrating the effectiveness of its step-wise modeling for sleep prediction.

Answer to RQ3. Table II, (b) evaluates the impact of long-term personalized features. *SOLAR* without long-term features results in a Log-Loss of 0.5739, whereas *SOLAR* with long-term features achieves 0.5560. This confirms that accumulated personal history effectively captures stable lifestyle patterns and sleep-related baselines that are not fully explained by short-term sensor observations.

Answer to RQ4. To answer RQ4, we analyze long- and short-term feature contributions from two complementary perspectives: group-level attribution and individual-feature explanations. First, we conduct a SHAP (SHapley Additive exPlanations)-based analysis [10], [11] to quantify feature-

TABLE III

SHAP-BASED FEATURE-GROUP ANALYSIS. LONG- AND SHORT-TERM GROUPS CONTAIN 377 AND 7,782 FEATURES, RESPECTIVELY; DENSITY IS NORMALIZED BY FEATURE COUNT.

Target	SHAP Share (%)		SHAP Density ($\times 10^{-4}$)	
	Long-term	Short-term	Long-term	Short-term
Q1	1.65	49.89	0.52	0.77
Q2	5.41	68.06	1.79	1.09
Q3	2.20	60.79	0.69	0.92
S1	11.64	37.31	4.14	0.64
S2	16.51	50.18	5.97	0.88
S3	15.99	40.01	5.68	0.69
S4	18.80	53.40	6.13	0.84

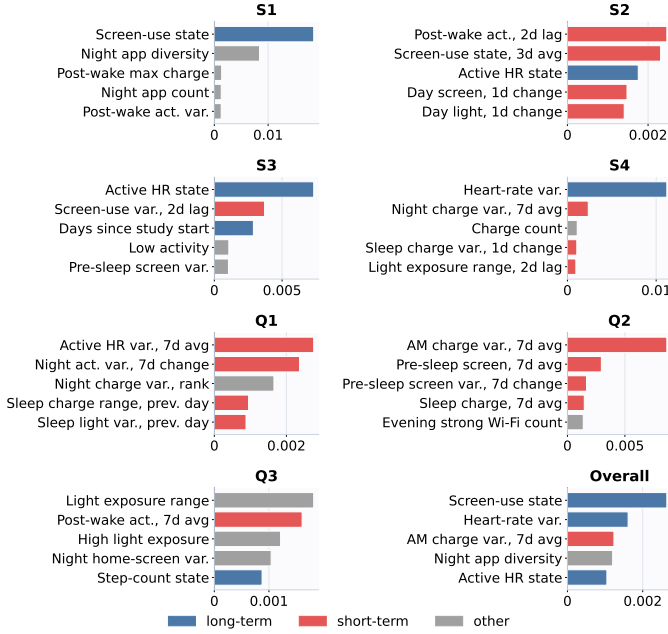


Fig. 3. Top 5 LIME feature importance patterns for each target and overall across long-term, short-term, and other feature groups.

group contributions. For each feature, we compute its mean absolute XGBoost SHAP value across all samples and aggregate these values within each feature group. We report two group-level metrics: SHAP Share, which denotes each group’s proportion of total mean absolute SHAP attribution, and SHAP Density, which denotes the average attribution magnitude per feature. As shown in Table III, the gap between long- and short-term attribution densities is relatively small for the Q-series targets. In contrast, the S-series targets show much larger gaps, with long-term attribution density consistently exceeding short-term density. This indicates that long-term behavioral and physiological patterns are particularly informative for predicting objective sleep outcomes.

We further examine the top-5 XGBoost features for each target using LIME (Local Interpretable Model-agnostic Explanations) [12]. While SHAP provides group-level attribution, LIME illustrates which individual features appear as salient evidence in local explanations. As shown in Fig. 3, long-term state-related features appear among the top-ranked ex-

planations across several targets, especially for the S-series targets. This finding shows that individual long-term features also provide salient evidence for specific predictions.

Together, the SHAP and LIME analysis suggest that accumulated personal history provides meaningful predictive signals, particularly for objective sleep indicators, while complementing recent day-specific observations.

V. CONCLUSION

We propose *SOLAR*, a stepwise framework for multimodal lifelog-based sleep prediction. *SOLAR* constructs long-term personalized features from behavioral histories to capture individual patterns and temporal trends. Based on these features, the framework anchors predictions using long-term states and then refines them with day-specific sensor measurements, day-specific variations, target-specific calibration, and temporal-neighbor priors. Experimental results show that *SOLAR* outperforms baselines as well as tabular backbone models and ensemble baselines in terms of Log-Loss. Ablation studies demonstrate that combining personalized long-term information with dynamic day-specific changes improves sleep-related prediction performance across multiple targets.

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